Simplifications of Context-Free Grammars

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A Substitution Rule

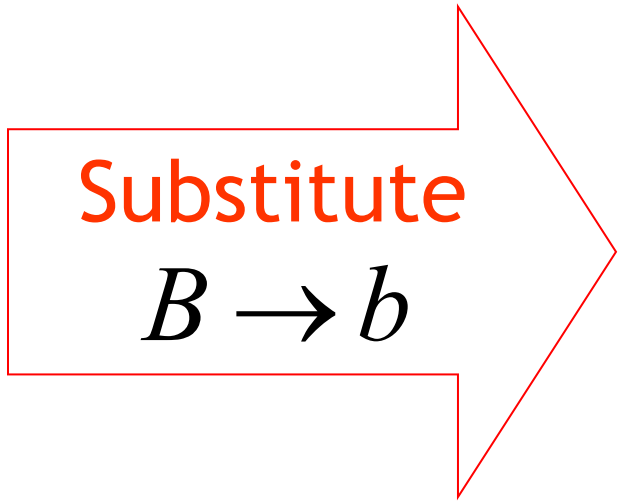
$S \to aB$

$A \to aaA$

$A \to abBc$

$B \to aA$

$B \to b$



Equivalent grammar

$S \to aB \mid ab$

$A \to aaA$

$A \to abBc \mid abbc$

$B \to aA$

$$S \rightarrow aB \mid ab$$

$$A \rightarrow aaA$$

$$A \rightarrow abBc \mid abbc$$

$$B \rightarrow aA$$

Substitute

$$B \rightarrow aA$$

$$S \rightarrow \cancel{aB} \mid ab \mid aaA$$

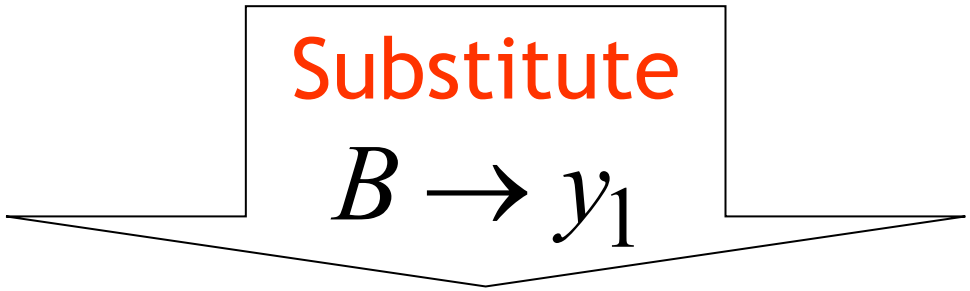
$$A \rightarrow aaA$$

$$A \rightarrow \cancel{abBc} \mid abbc \mid abaAc$$

Equivalent grammar

In general: $A \to xBz$

$B \to y_1$



$A \to xBz \mid xy_1z$

equivalent grammar

Nullable Variables

$\varepsilon - \text{production}: \quad X \to \varepsilon$

Nullable Variable: $Y \Rightarrow \ldots \Rightarrow \varepsilon$

Example:

$S \to aMb$

$M \to aMb$

$M \to \varepsilon$

Nullable variable

$\varepsilon - \text{production}$

Removing $\varepsilon-$productions

$S \to aMb$

$M \to aMb$

~~$M \to \varepsilon$~~

Substitute $M \to \varepsilon$

$S \to aMb \mid ab$

$M \to aMb \mid ab$

After we remove all the $\varepsilon-$productions all the nullable variables disappear. (except for the start variable.)

Unit-Productions

Unit Production: $X \to Y$

(A single variable in both sides)

Example:

$S \to aA$

$A \to a$

$A \to B$

$B \to A$

Unit Productions

$B \to bb$

Removal of unit productions:

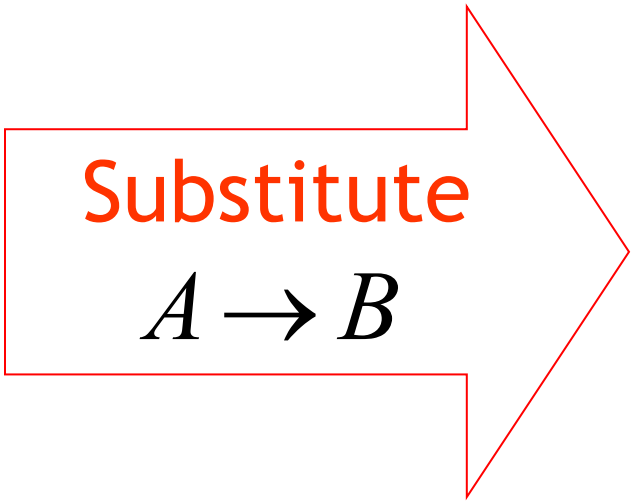
$S \to aA$

$A \to a$

~~$A \to B$~~

$B \to A$

$B \to bb$



$S \to aA \mid aB$

$A \to a$

$B \to A \mid B$

$B \to bb$

Unit productions of form $X \to X$

can be removed immediately.

$S \to aA \mid aB$

$A \to a$

$B \to A \mid B$

$B \to bb$

Remove

$B \to B$

$S \to aA \mid aB$

$A \to a$

$B \to A$

$B \to bb$

$S \to aA \mid aB$

$A \to a$

~~$B \to A$~~

$B \to bb$

Substitute

$B \to A$

$S \to aA \mid aB \mid aA$

$A \to a$

$B \to bb$

Remove repeated productions

$S \rightarrow aA \mid aB \mid aA$

$A \rightarrow a$

$B \rightarrow bb$



Final grammar

$S \rightarrow aA \mid aB$

$A \rightarrow a$

$B \rightarrow bb$

Useless Productions

$$S \to aSb$$

$$S \to \varepsilon$$

$$S \to A$$

$$A \to aA$$ Useless Production

Some derivations never terminate...

$$S \Rightarrow A \Rightarrow aA \Rightarrow aaA \Rightarrow \ldots \Rightarrow aa\ldots aA \Rightarrow \ldots$$

Another grammar:

$$S \to A$$

$$A \to aA$$

$$A \to \varepsilon$$

$$B \to bA$$ Useless Production

Not reachable from S

In general:

If there is a derivation

$$S \Rightarrow \ldots \Rightarrow xAy \Rightarrow \ldots \Rightarrow w \quad \in L(G)$$

consists of terminals

Then, variable A is useful.

Otherwise, variable A is useless.

A production $A \to x$ is useless

if any of its variables is useless

$S \to aSb$

$S \to \varepsilon$

Variables	Productions
	$S \to A$ useless
useless	$A \to aA$ useless
useless	$B \to C$ useless
useless	$C \to D$ useless

Removing Useless Variables and Productions

Example Grammar:

$$S \to aS \mid A \mid C$$

$$A \to a$$

$$B \to aa$$

$$C \to aCb$$

First: Find all variables that can produce strings with only terminals or ε

(possible useful variables)

$S \rightarrow aS \mid A \mid C$

$A \rightarrow a$

$B \rightarrow aa$

$C \rightarrow aCb$

Round 1: $\{A, B\}$

(the right hand side of production that has only terminals)

Round 2: $\{A, B, S\}$

(the right hand side of a production has terminals and variables of previous round)

This process can be generalized

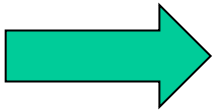
Then, remove productions that use variables other than $\{A, B, S\}$

$$S \rightarrow aS \mid A \mid C$$

$$A \rightarrow a$$

$$B \rightarrow aa$$

$$C \rightarrow aCb$$



$$S \rightarrow aS \mid A$$

$$A \rightarrow a$$

$$B \rightarrow aa$$

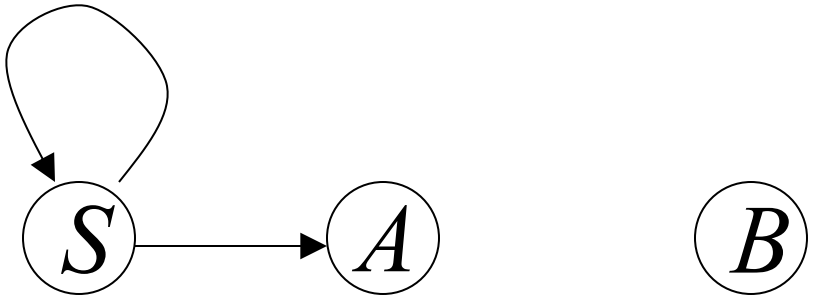
Second: Find all variables reachable from S

Use a Dependency Graph where nodes are variables

$$S \to aS \mid A$$

$$A \to a$$

$$B \to aa$$



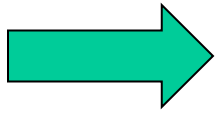
unreachable

Keep only the variables reachable from S

$S \rightarrow aS \mid A$

$A \rightarrow a$

~~$B \rightarrow aa$~~



Final Grammar

$S \rightarrow aS \mid A$

$A \rightarrow a$

Contains only useful variables

Removing All

Step 1: Remove Nullable Variables

Step 2: Remove Unit-Productions

Step 3: Remove Useless Variables

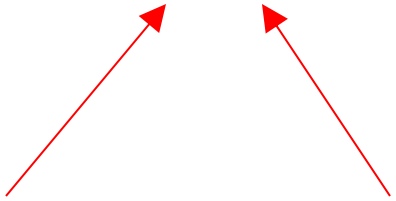
This sequence guarantees that unwanted variables and productions are removed.

Normal Forms
for
Context-free Grammars

Chomsky Normal Form

Each production has form:

$A \to BC$ or $A \to a$





variable variable terminal

(these may not be the start variables)

($S \to \varepsilon$ is allowed where S is the start variable)

Examples:

$$S \to AS$$
$$S \to a$$
$$A \to SA$$
$$A \to b$$

Chomsky Normal Form

$$S \to AS$$
$$S \to AAS$$
$$A \to SA$$
$$A \to aa$$

Not Chomsky Normal Form

Conversion to Chomsky Normal Form

Example:

$$S \to ABa$$

$$A \to aab$$

$$B \to Ac$$

Not Chomsky Normal Form

We will convert it to Chomsky Normal Form

Introduce new variables for the terminals:

$$T_a, T_b, T_c$$

$$S \to ABa$$

$$A \to aab$$

$$B \to Ac$$



$$S \to ABT_a$$

$$A \to T_a T_a T_b$$

$$B \to AT_c$$

$$T_a \to a$$

$$T_b \to b$$

$$T_c \to c$$

Introduce new intermediate variable V_1 to break first production:

$S \to ABT_a$

$A \to T_a T_a T_b$

$B \to AT_c$

$T_a \to a$

$T_b \to b$

$T_c \to c$



$S \to AV_1$

$V_1 \to BT_a$

$A \to T_a T_a T_b$

$B \to AT_c$

$T_a \to a$

$T_b \to b$

$T_c \to c$

Introduce intermediate variable: V_2

$$S \to AV_1$$
$$V_1 \to BT_a$$
$$A \to T_aT_aT_b$$
$$B \to AT_c$$
$$T_a \to a$$
$$T_b \to b$$
$$T_c \to c$$



$$S \to AV_1$$
$$V_1 \to BT_a$$
$$A \to T_aV_2$$
$$V_2 \to T_aT_b$$
$$B \to AT_c$$
$$T_a \to a$$
$$T_b \to b$$
$$T_c \to c$$

Final grammar in Chomsky Normal Form:

Initial grammar

$$S \to ABa$$

$$A \to aab$$

$$B \to Ac$$

$$S \to AV_1$$

$$V_1 \to BT_a$$

$$A \to T_aV_2$$

$$V_2 \to T_aT_b$$

$$B \to AT_c$$

$$T_a \to a$$

$$T_b \to b$$

$$T_c \to c$$

In general:

From any context-free grammar
not in Chomsky Normal Form,
we can obtain:

an equivalent grammar
in Chomsky Normal Form

The Procedure

First remove:

Nullable variables

Unit productions

(Useless variables optional)

Then, for every symbol a :

New variable: T_a

Add production $T_a \to a$

In productions with length at least 2

replace a with T_a

Productions of form $A \to a$

do not need to change!

Replace any production $A \to C_1 C_2 \cdots C_n$

with $A \to C_1 V_1$

$V_1 \to C_2 V_2$

$\cdots$

$V_{n-2} \to C_{n-1} C_n$

New intermediate variables: $V_1, V_2, \ldots, V_{n-2}$

Observations

- Chomsky normal forms are good for parsing and proving theorems.

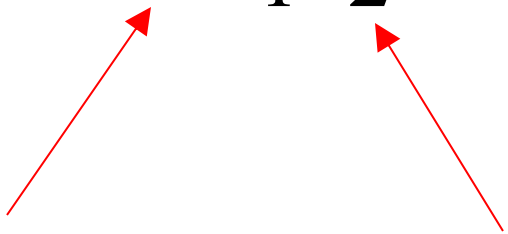
- It is easy to find the Chomsky normal form for any context-free grammar.

ε

Greinbach Normal Form

All productions have form:

$$A \rightarrow a\, V_1 V_2 \cdots V_k \qquad k \geq 0$$



symbol variables

Examples:

$$S \to cAB$$
$$A \to aA \mid bB \mid b$$
$$B \to b$$

Greinbach
Normal Form

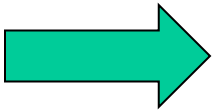
$$S \to abSb$$
$$S \to aa$$

Not Greinbach
Normal Form

Conversion to Greinbach Normal Form:

$$S \to abSb$$

$$S \to aa$$



$$S \to aT_bST_b$$

$$S \to aT_a$$

$$T_a \to a$$

$$T_b \to b$$

Greinbach Normal Form

Observations

- Greinbach normal forms are very good for parsing strings (better than Chomsky Normal Forms).

- However, it is difficult to find the Greinbach normal of a grammar.